

1. The ODE system $\begin{cases} y'(t) = f(t, y(t)) \\ y(t_n) = y_n \end{cases} \Rightarrow y(t_{n+1}) = y(t_n) + \int_{t_n}^{t_{n+1}} f(s, y(s)) ds$

Notice that $T_{n+1} = \frac{y_{n+1} - y_n}{h} - \Phi(t_n, y_n; h)$, $\Phi(t_n, y_n; h) = \frac{1}{2} [f(t_n, y_n) + f(t_{n+1}, y_{n+1})]$

Consider $\frac{d}{dt} f(t, y(t)) = f_t(t, y(t)) + f_y(t, y(t)) \cdot y'(t) = f_t + f_y \cdot f$ (which $f = y' = f(t, y(t))$)

and $\int_{t_n}^{t_{n+1}} f(s, y(s)) ds = hf_n + \frac{1}{2} h^2 (f_t + f_y f)_n + \frac{1}{6} h^3 \left(\frac{d^2}{dt^2} f(t, y(t)) \right)_n + O(h^4)$.

Hence for exact sol. in Taylor expansion: $y_{n+1} = y_n + h y'(t_n) + \frac{1}{2} h^2 y''(t_n) + O(h^3)$

$\Rightarrow y_{n+1} - y_n = hf_n + \frac{1}{2} h^2 (f_t + f_y f)_n + O(h^3)$

Then, consider $f(t_{n+1}, y_{n+1}) = f_{n+1} - f_y(t_{n+1}, y_{n+1})(y_{n+1} - (y_n + hf_n)) + O((y_{n+1} - (y_n + hf_n))^2)$

$\therefore y_{n+1} - (y_n + hf_n) = \frac{h^2}{2} (f_t + f_y f)_n + O(h^3)$

$\therefore f(t_{n+1}, y_{n+1}) = f_{n+1} - f_y(t_{n+1}, y_{n+1}) \cdot \frac{1}{2} h^2 (f_t + f_y f)_n + O(h^3)$

$\therefore \Phi(t_n, y_n; h) = \frac{1}{2} (f_n + f_{n+1}) + O(h^2) \Rightarrow h\Phi(t_n, y_n; h) = \frac{h}{2} (f_n + f_{n+1}) + O(h^3)$

conclude that, we get $E_1 = \int_{t_n}^{t_{n+1}} f(s, y(s)) ds - \frac{h}{2} (f_n + f_{n+1}) = -\frac{h^3}{12} \left(\frac{d^2}{dt^2} f \right)_n + O(h^4)$ by the error for the trapezoid rule in ch 9.

and $E_2 = \frac{h}{2} [f(t_{n+1}, y_{n+1}) - f(t_{n+1}, y_n + hf_n)] = \frac{h}{2} O(h^2) = O(h^3)$

$\therefore hT_{n+1} = E_1 + E_2 = O(h^3) \Rightarrow T_{n+1} = O(h^2)$, hence for $h \rightarrow 0$, $\tau(h) = O(h^2)$

$\therefore$ Heun's method has order 2 w.r.t h # (take $\tau(h) = \max_n |T_{n+1}(h)|$)

2. By the error of trapezoidal method in ch 9. $\Rightarrow E_1(f) = -\frac{h^3}{12} f''(\xi)$, $h = b - a$.

We can get a suitable $\xi_n \in (t_n, t_{n+1})$ s.t. $y_{n+1} = y_n + \frac{h}{2} [f_n + f_{n+1}] - \frac{h^3}{12} f''(\xi_n, y(\xi_n))$

$\therefore \frac{y_{n+1} - y_n}{h} = \frac{1}{2} [f_n + f_{n+1}] - \frac{h^2}{12} f''(\xi_n, y(\xi_n))$. By the trapezoidal method's relation,

$\therefore \frac{y_{n+1} - y_n}{h} = \frac{1}{2} [f_n + f_{n+1}]$, $\therefore$ we get $T_{n+1}(h) = O(h^2)$ (comparing $\frac{y_{n+1} - y_n}{2} - \frac{1}{2} f_n$)

Hence take $\tau(h) = \max_n |T_{n+1}(h)|$, $\tau(h) = O(h^2)$ as $h \rightarrow 0$, $\therefore$ the Crank-Nicolson method has order 2 w.r.t h #